

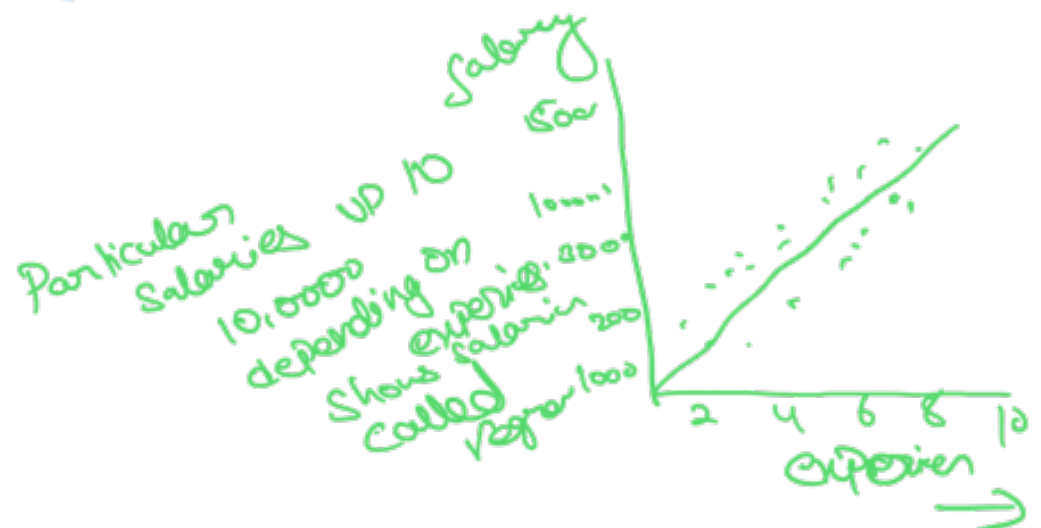
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P. P. S. S. S.

→ Regression analysis is a statistical method to model the relationship b/w a dependent and independent variables with one or more independent variables.

→ It helps us to understand how the value of the dependent variable is changing corresponding to an independent variable when other independent variables are held fixed.

→ It Predicts continuous/real values such as temperature, age, Salary, Price etc.



Terminology related to Regression analysis

① Dependent variable:-

The main factor in Regression analysis which we want to Predict or understand is called the dependent variable. It is also called target variable.

② Independent variable:-

The factors which affect the dependent variables or which are used to Predict the values of the dependent variables are called independent variable also called as a Predictor.

③ Outliers:- Outliers is an observation which contains either very low value or very high value in comparison to other observed values. An outlier may hamper the result, so it should be avoided.

Q. 1. ✓

so it should be avoided.

(i) Underfitting and overfitting:-

If our algorithm works well with the training dataset but not well with test dataset, then such problem is called Overfitting.

And if our algorithm does not perform well even with training dataset, then such problem is called Underfitting.



High test error
High training error



low test error
low training error



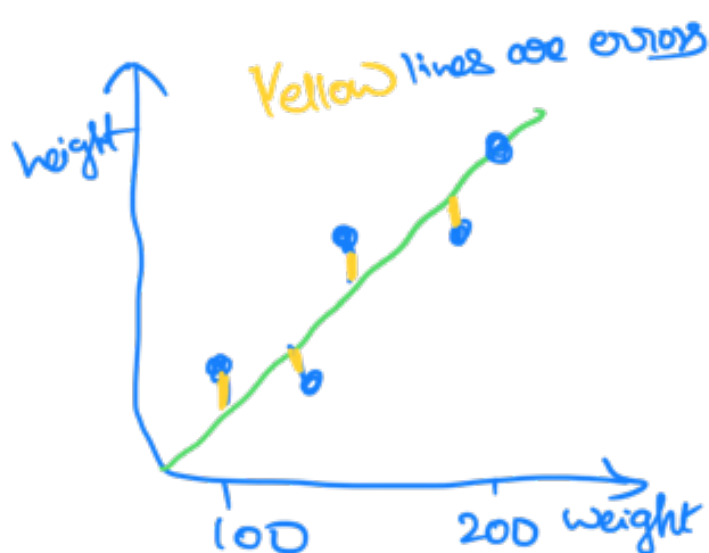
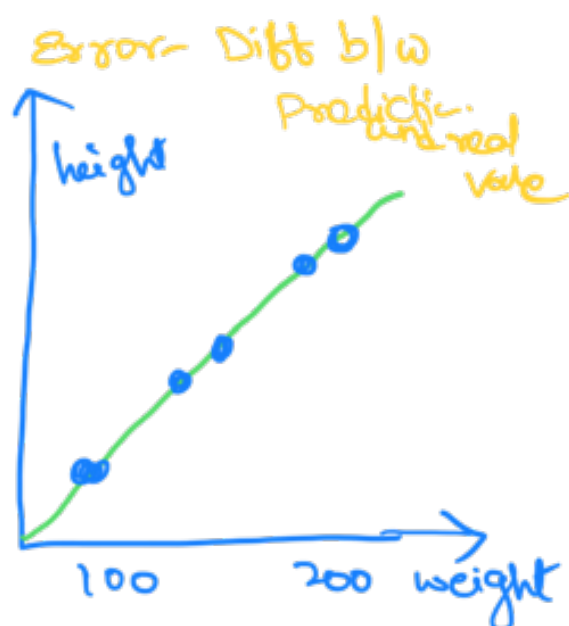
Low training error
High test error

→ Optimum is proper sol.

Derivation of Least Squares

Least Square method — To find the best fit line and minimize the distance b/w actual value and predicted value

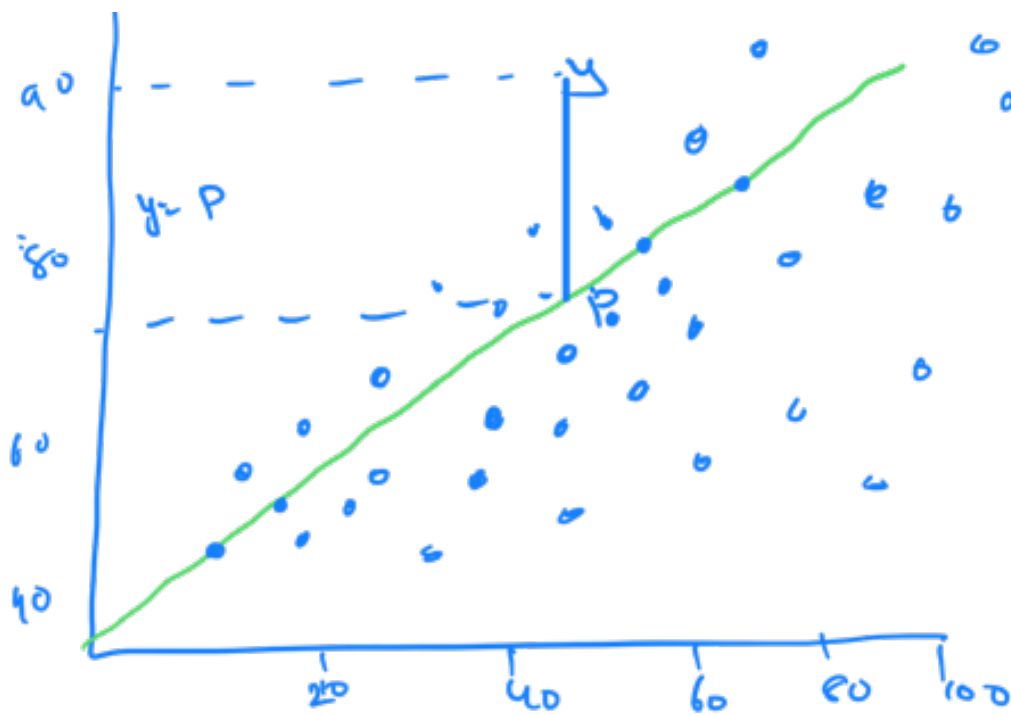
The least square regression method is a technique commonly used in Regression analysis. It is a Mathematical method used to find the best fit line that represents the relationship b/w an independent and dependent variable in such a way that the error is minimized.



→ All the data points are present on the particular decision line

⇒ Perfect accuracy is there
⇒ best fit line is there

Here there is only one data point lie in the best fit line and some are near to best fit line



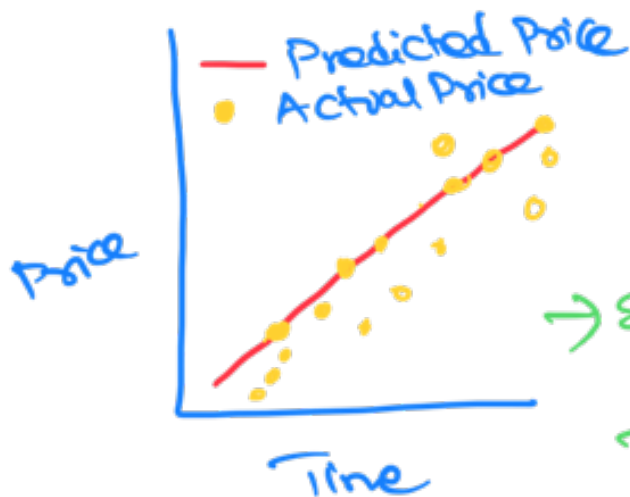
The dist b/w actual value and Predicted value are called errors.

To minimize this we use least squares reg method.

→ The line of best fit line is drawn across a scatter plot of data points in order to represent a relationship b/w those data points.

→ The Least square method is one of the most effective ways used to draw the line of the best fit. It is based on the idea that the square of the errors obtained must be minimized to the most possible extent and hence the name least square method

P. R. B. S.



As Time $\uparrow$ the Price $\uparrow$

→ Some dots are present on the best fit line. They show the accuracy 100%.

But some dots are near to predicted price which are errors i.e. Particular errors are present.

The work of regression is to find out the best fit line.

Try to cover maximum data points on the one line $\Rightarrow$ line of Best fit.

Steps to calculate the line of Best fit

Slope of line $\rightarrow$

$$y = mx + c$$

y — Dependent Variable
 x — Independent Variable
 m — Slope of the line
 c — Y-intercept

P. Pindoria

To calculate best fit line.

To start a construction of line
i.e. relationship b/w variables and
data points

- ① The slope of a line characterizes the direction of a line. To find the slope, you divide the difference of the y-coordinates of 2 points on a line by the difference of the x-coordinates of those same 2 pts.

$$m = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$m \rightarrow$ slope of line

$n \rightarrow$ total number of data points

$x \rightarrow$ Independent variable

$y \rightarrow$ Dependent variable



$$y = mx + c$$



From this fig: there is a regression line on x -axis $\rightarrow$ independent variable
on y -axis $\rightarrow$ Dependent variable

There is a regression line i.e. best fit line

This best fit line Join y -axis
 $\hookrightarrow y$ -intercept
denoted as c

$\rightarrow$  $\rightarrow$ are the data points.

Some are on best fit line and
some are on just near to best
fit line.

So the dist b/w actual value
and predicted value

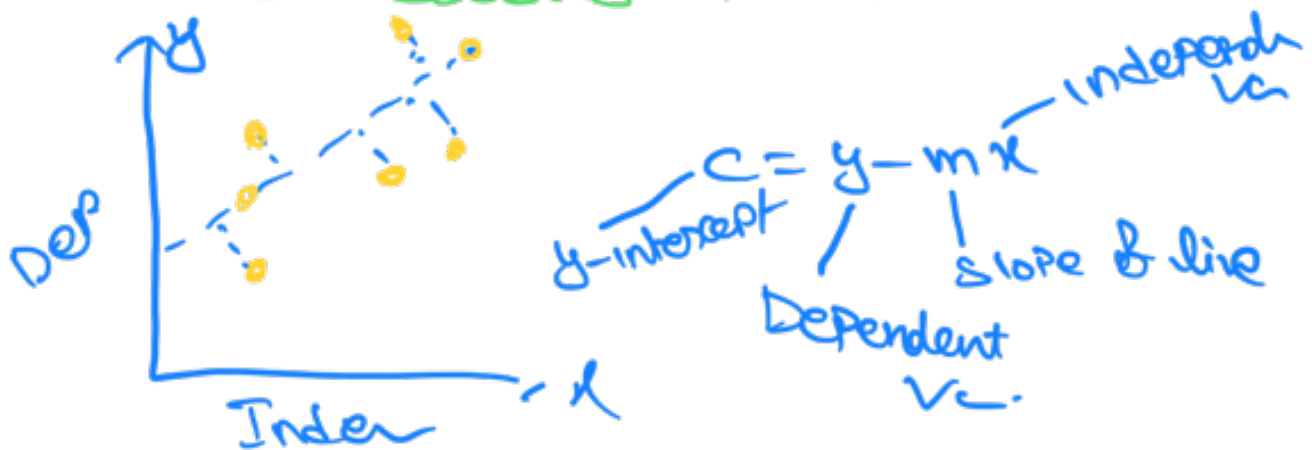
Dist b/w particular dotted line
and the yellow data dots is
called slope and denoted by m

① Find the value m by using least sq

② Calculate y value $C = y - mx$

Compute the y -intercept

The y -intercept of a line is the value of y at the point where the line crosses the y -axis



③ Substitute all values in $y = mx + C$.

Substitute all the values in the final eq

④ Simple eq that represents a st. line along 2-Dim data

i.e x-axis and y-axis

$$y = mx + c$$

Diagram showing the components of the linear equation $y = mx + c$:

- y is labeled as "Dependent Var".
- m is labeled as "Slope" and "B-line".
- x is labeled as "Independent Var".
- c is labeled as "y-intercept".

Example :-

→ Tom who is the owner of a retail shop, found the Price of different T-shirts vs the number of T-shirts sold at his shop over a period of one week.

- From the concept of least squares regression to find the line of best fit for the below data



Step 1 :- Calculate the slope 'm' x axis by using the following formula.

$$m = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

Price of T-shirt in dollars (x)	Number of T-shirts sold (y)
2	4
3	5
5	7
7	10
9	15

on substituting the respective values,
we get $m = 1.518$ approximately

Step 2:- Compute the y-intercept value
($m = 1.518$ as per step 1)

After substitute the respective value,
 $C = 0.305$ approximately

$$C = y - mx$$

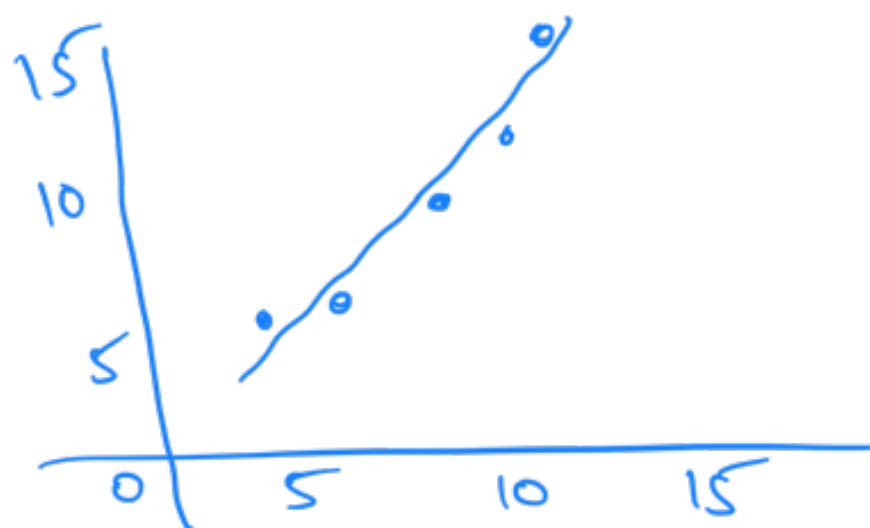
$$C = \frac{\sum y - m \sum x}{n}$$

Step 3:-

Substitute the values in the final eq. $\rightarrow y = mx + c$

Price of T-shirts in dollars (x)	Number of T-shirts sold (y)	<u>$y = mx + c$</u>	error
2	4	3.3	-0.67
3	5	4.9	-0.14
5	7	7.9	0.89
7	10	10.9	0.93
9	15	13.9	-1.03

Graph that represents the $y = mx + c$ line of best fit



error
 $\downarrow$
dist b/w
actual value
and pred
 $y = 4$ Pred = 3.3
 $4 - 3.3 = 0.67$

→ Now Tom can use the above eq

to estimate how many T-shirts

of Price \$8 Can be sell at the retail shop.

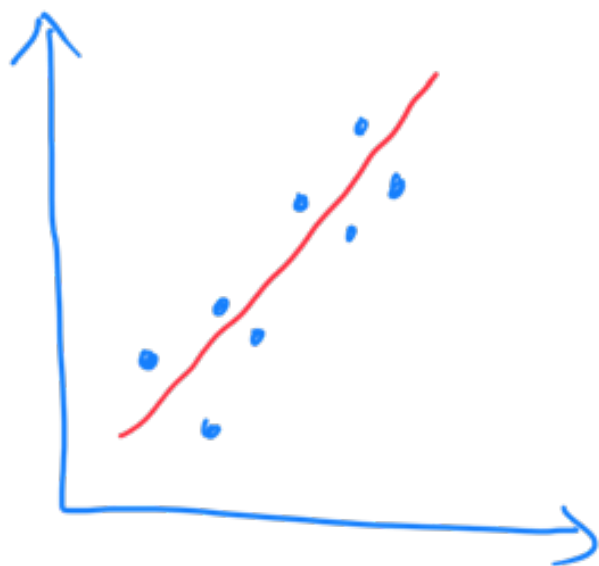
$$y = mx + c$$
$$y = 1.518 \times 8 + 0.305$$
$$= 12.45 \text{ T-shirts}$$

→ This comes down to 13 T-shirts

→ Write a linear equation that "best fits" the data in the table shown below.

x	1	2	3	4	5	6	7
y	1.5	3.8	6.7	9.0	11.2	13.6	16

Sol:-



all pts are
as close to as
close to that line
Using the least sq
method we can
find the eq of this
line

To test the eq to see how
accurate it is in given y values
to input the x values.

x	y	xy	x ²
1	1.5	1.5	1
2	3.8	7.6	4
3	6.7	20.1	9
4	9.0	36	16
5	11.2	56	25
6	13.6	81.6	36
7	16	112	49
$\Sigma x = 28$ $\Sigma y = 61.8$ $\Sigma xy = 314.8$ $\Sigma x^2 = 140$			

$$y = mx + c$$

$$y = c_0 + c_1 x$$

$$\text{Slope} = m = b_1$$

$$y\text{-int} = b = b_0$$

$$y = mx + c$$

$$m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{7(314.8) - (28)(61.8)}{7(140) - (28)^2}$$

$$m = 2.4142857$$



$$c = \frac{\sum y - m \sum x}{n}$$

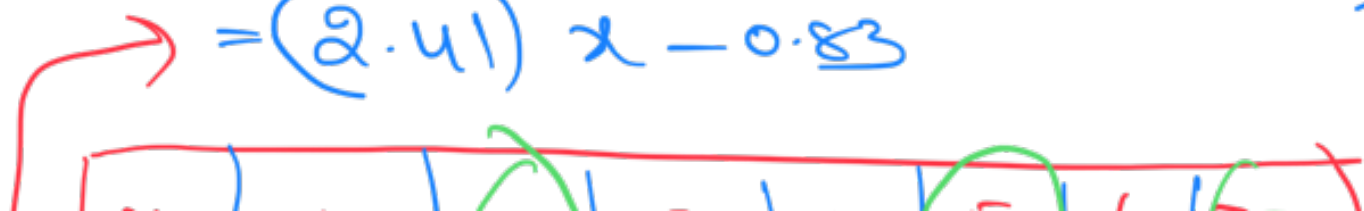
$$= \frac{61.8 - (2.4142857)(28)}{7}$$

$$= -0.828571$$

$$y = mx + c$$

$$y = (2.4142857)x + (-0.828571)$$

$$= (2.41)x - 0.83$$



x	1	2	3	4	5	6	7
y	1.5	3.8	6.7	9.0	11.2	13.6	16

$$y = 2.41(2) - 0.83$$

$$\Rightarrow 3.99$$

(At $x=2$
 $y=3.99$ getting
 it is not too far
 from 3.8
 $\Rightarrow$ Good approx

$$y = 2.41(5) - 0.83 = 11.22 \text{ close to } 11.2$$

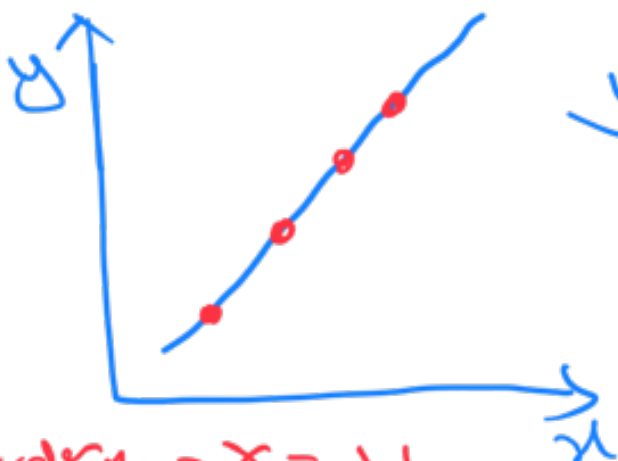
$$y = 2.41(7) - 0.83 = 16.04 \text{ close to } 16$$

This linear Eq fits the observed data very well

$\rightarrow$ Calculate the correlation coeff
 b/w the two variables
 x and y shown below

x	1	2	3	4	5	6
y	2	4	7	9	12	14

Q

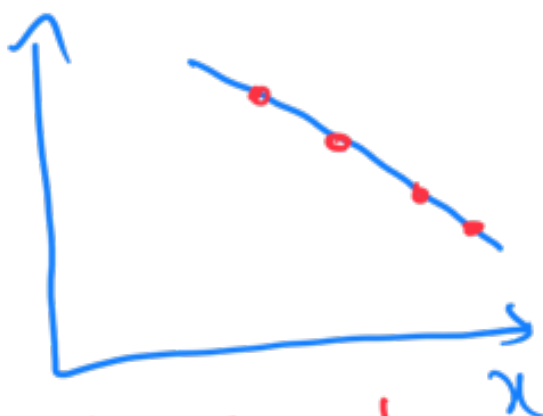


Correlation $\rightarrow r = +1$

all Pt on line
and Line has Positive
slope

Positive correlation
becoz when

as $x \uparrow$ y also $\uparrow$

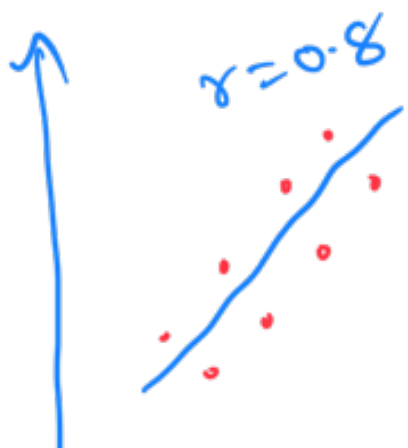


Correlation $\rightarrow r = -1$

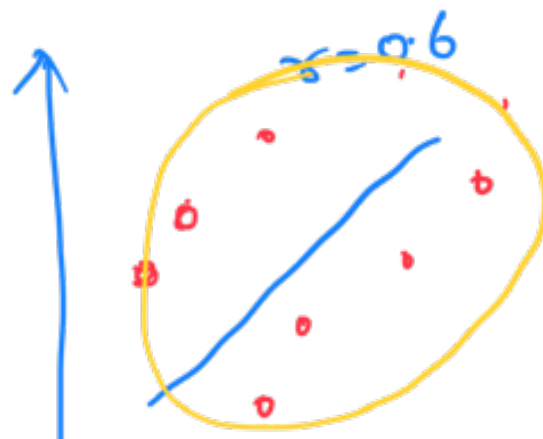
all the Pt on
line and
Line has Negative
slope

Negative correlation
becoz

as $x \uparrow$ $y \downarrow$



$r = 0.8$



$r = 0.6$

✓

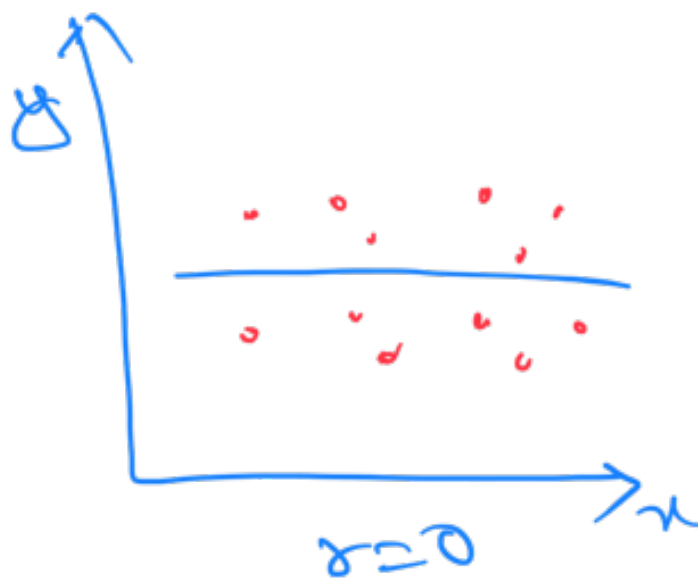
$$0 < r < 1$$

Line is $\uparrow$ and has
Positive slope

Lines are somewhere
b/w 0 and 1 but $r \neq 1$
might be $r = 0.8$
close to 1 than 0

$$r =$$

more scattered
 r value may
be less
but somewhere
b/w 0 and 1
 r is close
to 0 than 1



In this case
 r is very close to 0

P. D. B. S. ...

Correlation Coeff is the linear
relationship b/w two variables.

x	1	2	3	4	5	6
y	2	4	7	9	12	14

x	y	xy	x ²	y ²
1	2	2	1	4
2	4	8	4	16
3	7	21	9	49
4	9	36	16	81
5	12	60	25	144
6	14	84	36	196
$\Sigma x = 21$	$\Sigma y = 48$	$\Sigma xy = 211$	$\Sigma x^2 = 91$	$\Sigma y^2 = 490$

Correlation Coeff

$$\begin{aligned}
 r &= \frac{n \Sigma xy - \Sigma x \Sigma y}{\sqrt{[n \Sigma x^2 - (\Sigma x)^2] [n \Sigma y^2 - (\Sigma y)^2]}} \\
 &= \frac{6(211) - 21(48)}{\sqrt{[6 \cdot 91 - (21)^2] [6 \cdot 490 - (48)^2]}} \\
 &= \frac{1266 - 1008}{\sqrt{[546 - 441] [2940 - 2304]}}
 \end{aligned}$$



$$\frac{\sqrt{[546-441][2940-2304]}}{258}$$

$$= \frac{258}{\sqrt{105 \cdot 636}}$$

$$\boxed{r = 0.998}$$


 r is positive
 $\Rightarrow x \uparrow y \uparrow$
 Positive correlation

→ Determine the regression eq. of y on x
 by the method of least squares
 from the following table.

x	5	8	7	6	4
y	3	4	5	2	1

8

x	y	xy	x^2
-----	-----	------	-------

5	3	15	25
8	4	32	64
7	5	35	49
6	2	12	36
4	1	4	16
30	15	98	190



The regression eq of y on x
 $y = a + bx$

Subspace Formulation of linear models

Vector Subspace

A non-empty set V is called a

a) non empty set V is called a
 real (complex) vector space if in V
 are defined two algebraic operations
 vectors addition and scalar multiplication
 as follows

for $u, v \in V$

- 1) $u+v \in V$ (closure Property)
- 2) $u+v = v+u$ (Commutativity)
- 3) $(u+v)+w = u+(v+w)$ (Associativity)
- 4) $u+0 = u$ (Existence of unique zero element in V)
- 5) $u+(-u) = 0$ (Existence of additive inverse in V)

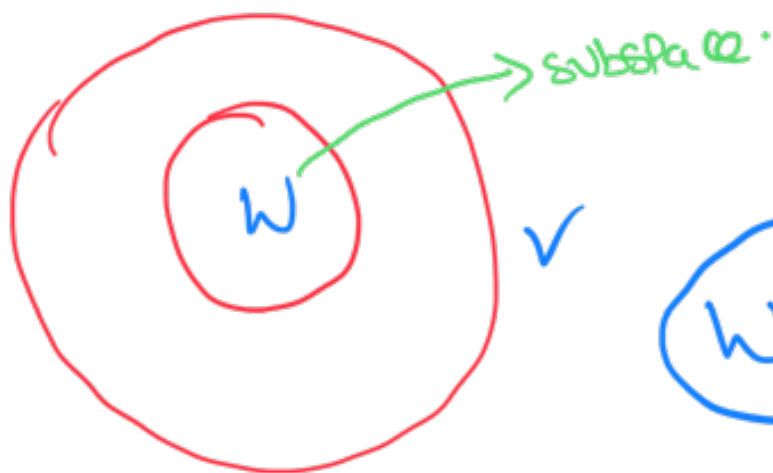
for any scalar d

- 1) $du \in V$ (closed under multiplication)
- 2) $d(u+v) = du + dv$ (distributivity)
- 3) $(\alpha+\beta)u = \alpha u + \beta u$ (distributivity)
- 4) $d(\beta u) = (d\beta)u$ (associativity)
- 5) $1u = u$ (Existence of multiplicative inverse in V)

(*) If the elements V (i.e. $u, v, w, \dots$)

and scalars $\alpha, \beta, \dots$ are real then V is called real vector space otherwise V is called a Complex vector space.

Subspace



P. P. Subudhi

A non-empty subset W of a vector space V is said to be subspace of V if W is vector space under the same operations of addition and scalar multiplication as in V

(or)

A non-empty subset W of a vector space V is said to be subspace of V iff for $u, v \in W$ and $\alpha, \beta \in \mathbb{R}$

$$\alpha u + \beta v \in W$$



The necessary and sufficient conditions
for W be a subspace of V

for $u, v \in W$ and scalar α

- ① $0 \in W$
- ② $u+v \in W$
- ③ $\alpha u \in W$



ex Let V be a v.s in $\mathbb{R}^3$.

Check whether it is subspace or not

① $W = \{(a, b, c) \mid a \leq 0\}$

1

Let $u = (a, b, c) \in W$
 $\Rightarrow a \leq 0$

Let $\alpha u = (\alpha a, \alpha b, \alpha c)$

If $\alpha = -4 \in \mathbb{R}$ ^{not always ≤ 0} $\times$ can't say.

If $\alpha u = (-4a, -4b, -4c) \notin W$

⊗

Because $-4a \not\leq 0$ ^{not always} as $a \leq 0$

Hence, W is not subspace.

② $W = \{(a, b, c) \mid c \text{ is integer}\}$

§ Let $u = (a, b, c) \in W$
 $\Rightarrow c \in \mathbb{Z}$

$$\alpha u = (\alpha a, \alpha b, \alpha c)$$

$$\alpha = \sqrt{2} \in \mathbb{R}$$

Then

$$\alpha u = (\sqrt{2}a, \sqrt{2}b, \sqrt{2}c) \notin W$$

because $\sqrt{2}c \notin \mathbb{Z}$

Hence, W is not subspace.

Q Let V be a vector space in $\mathbb{R}^3$.
Examine whether the following are the
subspaces or not?

$$(i) W = \{ (a, b, c) \mid a^2 + b^2 + c^2 \leq 1 \}$$

§ Let $u = (a, b, c) \in W$
 $\Rightarrow a^2 + b^2 + c^2 \leq 1$

To check whether

for $u, v \in W$ and $\alpha \in \mathbb{R}$

$$u + v \in W$$

$$\alpha u \in W$$



let $v = (a_2, b_2, c_2)$

$du = (da, db, dc)$

if $da = 3 \in \mathbb{R}$

Then $du = (3a, 3b, 3c)$

$9a^2 + 9b^2 + 9c^2 \leq 9$

can also be written as $9(a^2 + b^2 + c^2) \leq 9 \neq 1$ never!

< 9 it can be 7.8 or 8....

Hence W is not subspace

② $W = \{(a, b, c) \mid a=b=c\}$

Let $u = (a_1, b_1, c_1) \in W$

$v = (a_2, b_2, c_2) \in W$

$\Rightarrow a_1 = b_1 = c_1$ and $a_2 = b_2 = c_2$

Now $u + v = (a_1 + a_2, b_1 + b_2, c_1 + c_2)$

Since $a_1 = b_1 = c_1$; $a_2 = b_2 = c_2$

$\Rightarrow a_1 + a_2 = b_1 + b_2 = c_1 + c_2$

Thus $u + v \in W$ ✓



For scalar α , we have

$$\alpha u = (\alpha a_1, \alpha b_1, \alpha c_1)$$

$$\text{as } a_1 = b_1 = c_1 \Rightarrow \alpha a_1 = \alpha b_1 = \alpha c_1$$

Thus $\alpha u \in W$ ✓

Hence W is a subspace of V .

Take $\alpha, \beta \in \mathbb{R}$ (or)

$$\text{then } \alpha u + \beta v = (\alpha a_1 + \beta a_2, \alpha b_1 + \beta b_2, \alpha c_1 + \beta c_2)$$

To prove

$$\alpha a_1 + \beta a_2 = \alpha b_1 + \beta b_2 = \alpha c_1 + \beta c_2$$

$$\text{Since } a_1 = b_1 = c_1 ; a_2 = b_2 = c_2$$

$$\Rightarrow \alpha a_1 + \beta a_2 = \alpha b_1 + \beta b_2 = \alpha c_1 + \beta c_2$$

Hence, W is a subspace of V .

Q Check whether the following W is subspace of $V = \mathbb{R}^3$ or not.

$$W = \{ (x, y, z) / x + y - z = 0 \}$$

8

Take $u, v \in W$ such that
 $u = (1, 0, 1), v = (0, 1, 1)$

$0+0 \neq 1$
 $1+1 = 1$

$0 \notin W$

Then $u+v = (1, 1, 2) \notin W$
because $1+1 \neq 1$

Take $d=2$ then

$$du = (2, 0, 2) \notin W$$

because $2+0 \neq 1$

Hence W is not subspace

→ Check whether the following W is
Subspace of $V = \mathbb{R}^3$ or not?

$$W = \{(x, y, z) \mid xy = 0\}$$

8

Take $u, v \in W$ such that

$$u = (1, 0, 1), v = (0, 1, 1)$$

$$u+v = (1, 1, 2) \notin W$$

because $1 \cdot 1 \neq 0$

$\therefore W$ is not subspace.

Q. Baird & Co.

Orthogonal Projections

Orthogonal Complement :- ^{perp}

The set of all vectors that are orthogonal to w

If W is a subspace of finite dimensional inner product space V , then $V = W \oplus W^\perp$

$$w^{\perp\perp} = w$$

*

dem v. dem mit dem ist

Def:- Orthogonal Projection

If W is a subspace of a finite dimensional inner product space V , then $V = W \oplus W^\perp$.

Therefore, every vector $\alpha \in V$ can be uniquely expressed as $\alpha = \alpha_1 + \alpha_2$, where $\alpha_1 \in W$ and $\alpha_2 \in W^\perp$.

The vectors α_1 and α_2 are called the orthogonal projections of α on the subspaces W and $W^\perp$.

Orthogonal complement:- The set of all vectors that are orthogonal to W is called the orthogonal complement of W and is denoted by $W^\perp$, $W^\perp$ is a subspace of V .

1) A vector x is in $W^\perp$ iff x is orthogonal to every vector in a set that spans W .

2) If S is any set of vectors in a

inner product space $V(F)$, then W is a subspace of V

exam:-

(Proj) $\rightarrow A(A^T A)^{-1} A^T \vec{x}$

$V = \text{span} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right)$

Basis of V

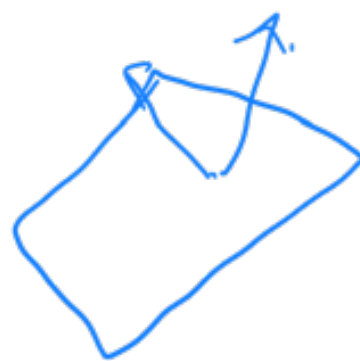
$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$

← Transposed

Proj = $\frac{1}{5} \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

find the O.P matrix onto plane

$x + y - z = 0$



↓

$P = A(A^T A)^{-1} A^T$

matrix Subspace we are projecting

$A = \begin{pmatrix} 1 & 1 \\ a_1 & a_2 \\ 1 & 1 \end{pmatrix} \quad a_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Basis for plane

$$a_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 0 & 1 \end{pmatrix} \quad A^T = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad (A^T A)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 0 & 1 \end{pmatrix} \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$P = \frac{1}{3} \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$



Projection $\rightarrow$ the shadow

So what is shadow of that vector
on any given plane

Let us consider the orthogonal
Projection of $x \in \mathbb{R}^3$ to a subspace
 $\mathbb{R}^3$

47. 17.

where $W \rightarrow x$ -axis

Projection of $x = [x \ y \ z]^T$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = (x \cdot e_1) e_1$$

Projection of
x-axis

↓
Standard
basis

$\omega \rightarrow x-y$ Plane

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

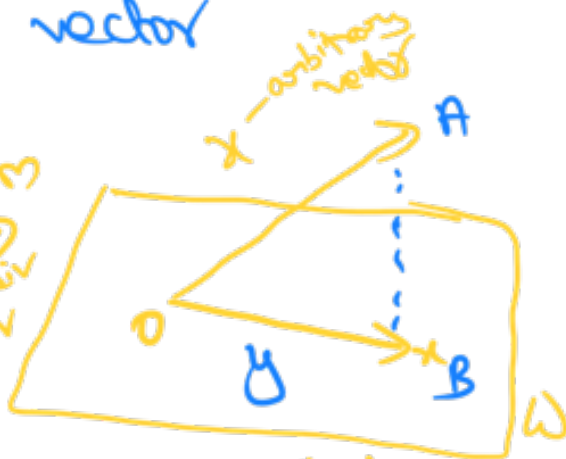
$$= (x_1 \cdot e_1)e_1 + (x_2 \cdot e_2)e_2$$

Def:- Let W be a linear subspace of $\mathbb{R}^n$ and $x \in \mathbb{R}^n$. A vector y is in W such that

$x - y \perp W$ is called **vector from**
subspace W **to this dir**

Orthogonal Projection
AB is normal to
eq triangle law
apply

⇒ This is the min of OA vector & OB vector



check by if we $OAB \Rightarrow OB + AB = OA$
 $AB = OA - OB = x - y$
 If $x - y$ is perpendicular to W $\Rightarrow x - y$ is perpendicular to W

(*) $x - y \in W^\perp$ (Perpendicular subspace)

Result :-

$$x - y \perp W \Leftrightarrow x - y \in W^\perp$$

Ex Let W be a non zero vector and $x \in \mathbb{R}^n$. Let y be the orthogonal projection of x to the line $\langle W \rangle$ spanned by W . Then y is characterized by $y \in W$ and $(x - y) \cdot W = 0$

Since $y \in \langle W \rangle$, $y = \sigma W$ for scalar $\sigma \in \mathbb{R}$

$$\begin{aligned} 0 &= (x - y) \cdot W = x \cdot W - y \cdot W \\ &= x \cdot W - \sigma W \cdot W \\ 0 &= x \cdot W - \sigma (W \cdot W) \\ \sigma (W \cdot W) &= x \cdot W \end{aligned}$$

y calcul becoz y is o.p of

$$\sigma = \frac{x \cdot W}{W \cdot W}$$

$$y = \sigma W$$

$$y = \frac{x \cdot W}{W \cdot W} W$$



$$0 = \frac{x \cdot w}{\|w\|^2} w$$

$$y = \frac{x \cdot w}{\|w\|^2} w$$

Uniqueness of O.P.:-

Thm:- Let W be a linear subspace of $\mathbb{R}^n$ and $x \in \mathbb{R}^n$. Then the O.P of x to W is unique.

Linear maps onto a subspace

W : Subspace of V

Is there a linear operator T s.t
range(T) = W ?

Yes... there are many
Basis for $V = \{v_1, v_2, \dots, v_n\}$

Define T as mapping each v_i to some $w_i \in W$
orthogonal projection:-

linear operator taking
 $v \in V$ into W in a specific way



Orthogonal Complementary:-

W : subspace of V

$W^\perp$: orthogonal complement of W

$$W^\perp = \{ v \in V : \langle v, w \rangle = 0 \ \forall \ w \in W \}$$

$$V = W \oplus W^\perp$$

How to find $W^\perp$?

Orthonormal basis for $W : \{ e_1, \dots, e_m \}$
Extend to orthonormal basis for

$$V : \{ e_1, \dots, e_m, e_{m+1}, \dots, e_n \}$$

Orthonormal basis for $W^\perp : \{ e_{m+1}, \dots, e_n \}$

$$v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m + \\ \langle v, e_{m+1} \rangle e_{m+1} + \dots + \langle v, e_n \rangle e_n$$

$$w = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m \in W$$

$$w^\perp = \langle v, e_{m+1} \rangle e_{m+1} + \dots + \langle v, e_n \rangle e_n$$

$$v = w + w^\perp$$

ex

$\mathbb{R}^4$

$$W = \text{Span} \{ (1, 0, 0, 0), (0, 1, 0, 0) \}$$

Subspace spanned by two vectors

orthonormal basis extension

$$\{ (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1) \}$$

$$W^\perp = \text{Span} \{ (0, 0, 1, 0), (0, 0, 0, 1) \}$$

$$v = (1, 0, 1, 1) = w + w^\perp$$

$$\text{where } w = (1, 0, 0, 0)$$

$$w^\perp = (0, 0, 1, 1)$$

$$W = \text{Span} \left\{ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0 \right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, 0 \right) \right\}$$

~~Terribly~~

— 0

All

the best